Yunus Emre Danabaş

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EDUCATION

Sabancı University Istanbul, Turkey
B.Sc. Double Major 2020–2026
CGPA: **3.77/4.00**

- **Mechatronics Engineering** (Graduated June 2025; *Ranked 1st in Department*)
- **Electronics Engineering** (Graduated June 2026; *Ranked 2nd in Department*)
- Combined specialization in robotics, control, and embedded systems across both majors
- Selected coursework: Kinematics and Dynamics of Machines; Introduction to Robotics; Autonomous Mobile Robotics; Deep Learning for Robot Control

RESEARCH & ENGINEERING EXPERIENCE

As Robotics (COMAU Turkey) Istanbul, Turkey
Part-time Robotics Engineer; Technical Lead, EVKA Wireless TÜBİTAK Project Oct 2025–Jul 2026

- Led the technical development of EVKA Wireless / EVKA 2.0, a TÜBİTAK-backed wireless tele-imitation system for industrial robot programming with COMAU robots
- Architected the embedded-to-ROS pipeline using ESP32-based sensing, dual-IMU/UWB positioning concepts, micro-ROS over WiFi, ROS 2 visualization, and PDL2-oriented robot path generation
- Developed EVKA Position as a separate encoder-based 3D positioning platform, converting rotary and draw-wire encoder data into Cartesian coordinates with calibration, filtering, dashboard, and TCP/WebSocket/serial interfaces
- Designed and fabricated capacitive grip/touch sensor PCBs using KiCad and LPKF, including AT42QT1011-based capacitive sensing hardware for industrial gripper applications
- Built C++/Python integration and testing tools while gaining hands-on experience with COMAU robot operation, PDL2 programming, and controller-level workflows

PRISMA Lab, University of Naples Federico II | `project page` Naples, Italy
Research Intern (Supervisors: Prof. Vincenzo Lippiello; Prof. Bruno Siciliano) Jun 2025–Oct 2025

- Defined a per-side actuated suspension concept for a compact lunar micro-rover, enabling ride height (heave) and roll leveling while preserving passive pitch averaging (rocker-like behavior)
- Initiated a ROS 2 Humble + Gazebo simulation pipeline (CAD → URDF/Xacro; `ros2_control` joints/controllers; posture presets via a Posture Manager) and handed off Gazebo C++ plugin development for passive differential emulation
- Synthesized design guidelines from lunar rover locomotion/suspension literature and actuator-integration constraints

MIRMI, Technical University of Munich (TUM) | `project page` Munich, Germany
Research Intern (Supervisors: M.Sc. Mehmet Can Yıldırım; Prof. Sami Haddadin) Jun 2024–Aug 2024

- Investigated crossed-flexure (cross-spring) stages for 6-DoF force–torque sensing, quantifying moment-capacity vs. force-sensitivity trade-offs
- Developed a parametric MATLAB modeling pipeline (dimensionless formulation + Newton–Raphson solver) to sweep geometries and load cases; estimated stiffness, parasitic motion, and strain
- Designed CAD and fabricated instrumented prototypes; built a MATLAB vision-metrology pipeline (ROI/edge detection/curve fitting) to track deformation and validate model predictions
- Created a Gazebo digital twin to reduce risk and accelerate controller development prior to hardware testing

The Scientific and Technological Research Council of Turkey (TÜBİTAK) | `project page` Gebze, Turkey
Robotics Engineering Intern Sep 2023–Oct 2023

- Completed an intensive ROS training program covering navigation, SLAM, and multi-robot coordination
- Built a Gazebo-based frontier exploration pipeline (`explore_lite` + `move_base`/DWA + gmapping) and benchmarked 1/2/3-robot configurations
- Integrated `multirobot_map_merge` and RViz to validate merged-map coverage and multi-agent performance

Pubinno Inc. Istanbul, Turkey
Mechanical Engineering Intern Jun 2023–Sep 2023

- Designed and 3D-printed prototypes in Onshape to accelerate iteration speed while meeting manufacturing constraints
- Built an automated maintenance prototype for first-line servicing; supported supplier QA/QC and documentation

TEACHING EXPERIENCE

Sabancı University Istanbul, Turkey
Teaching Assistant Sep 2024–Jun 2026
Instructors: Prof. Volkan Patoğlu, Asst. Prof. Dr. Meltem Elitaş, Asst. Prof. Melih Türkseven
Courses: ME312 Analysis and Synthesis of Mechanisms; ME403 Introduction to Robotics; ME304 Motion Control Systems

- Led labs/recitations, held weekly office hours, and supervised course projects for 15+ students per semester

PROJECTS

Hand-Steer Sim — Gesture Teleoperation for Mobile Robots | project page | code Mar 2025–May 2025

- Developed a camera-only teleoperation stack for differential-drive robots in ROS Noetic: MediaPipe hand landmarks → dual-branch TFLite inference (MLP for static commands, LSTM for steering trajectories) → `geometry_msgs/Twist`
- Achieved 99.65% accuracy (static) and 99.77% macro-F1 (dynamic) on a held-out test set; measured 13 ms end-to-end latency on an RTX 4060 Ti (~25 ms on laptop CPU) at 30 FPS
- Packaged a reproducible ROS+Gazebo workspace with one-command `roslaunch`; released a data-recording GUI, training notebooks, pretrained TFLite models, and CPU/GPU Dockerfiles

Passive Walker RL Pipeline (Graduation Project) | project page | code Oct 2024–May 2025

Supervisors: Prof. Volkan Patoğlu, Dr. Aykut Cihan Satici

- Trained a passive-dynamic biped via FSM → behavior cloning → PPO (JAX/Equinox/Optax; Brax/MuJoCo)
- Built expert controllers and a GPU-parallel PPO pipeline; achieved stable gait within $\sim 10^5$ steps
- Released reproducible code, configs, and logs; ran large hyperparameter sweeps for robust policies

SURover (Sabancı University Rover Team), Technical Team Captain Jun 2022–Jan 2025

- Built a ROS+Gazebo digital twin; authored URDF/Xacro and MoveIt pipelines for arm and drivetrain
- Implemented multi-sensor fusion and ArUco-based fiducial SLAM with ZED2, RealSense D435i, and RPLIDAR-A3
- Developed a Wi-Fi joystick and GUI with watchdog and E-stop; led mechanical fabrication/cabling and Arduino-based motor control for a field-ready rover

Cart-Pole Swing-Up Control (JAX & MuJoCo) | project page | code Oct 2024–Jan 2025

Supervisor: Dr. Aykut Cihan Satici

- LQR and neural controllers with differentiable closed-loop simulation (JAX/Diffrax/Equinox); trained NN policies via end-to-end backprop through ODE integration using energy-based loss and curriculum learning; validated robust swing-up in MuJoCo under external disturbances

Robotic Item Placement on Cluttered Desks (Baxter) | project page | code Jan 2023–Jan 2024

Supervisors: Prof. Volkan Patoğlu, Prof. Esra Erdem

- Ported the Baxter SDK to ROS Noetic (Python 3), repairing MoveIt integration and the Gazebo pipeline for sim-to-robot deployment
- Group project on Bayesian optimization with transfer learning for sequential, collision-free placement on cluttered surfaces; demonstrated clutter-aware placement policies

HONORS & AWARDS

- Merit-Based Full Tuition Scholarship — Sabancı University 2020–2025
- Dean's High Honor — Sabancı University 2020–2025
- Graduated 1st in the Mechatronics Engineering Department — Sabancı University 2025
- Graduated 2nd in the Electronics Engineering Department — Sabancı University 2026
- Extracurricular Student Activities Leadership Award, committee selection for SURover leadership 2023, 2024

TECHNICAL SKILLS & LANGUAGES

CAD & Fabrication:	SolidWorks (CSWA); Onshape; Fusion 360; FDM 3D printing
Robotics Stack:	ROS 1/2; micro-ROS; URDF/Xacro; Gazebo; `ros2_control`; MoveIt/MoveIt2; Navigation2
Modeling & Control:	MATLAB/Simulink; Autolev
Learning & Simulation:	JAX (Equinox, Optax, Diffrax); MuJoCo; Brax; RL (BC, PPO)
Perception:	OpenCV; MediaPipe; ArUco; RealSense SDK
Programming:	Python; C/C++; MATLAB; Assembly
Embedded & Electronics:	Arduino; ESP32; Verilog; LTspice; PCB design (KiCad); PCB prototyping (LPKF S63)
Tools & DevOps:	Git; Docker; Linux
Languages:	English (Advanced); Turkish (Native)

ACTIVITIES & VOLUNTEERING

Sabancı IEEE Student Chapter | Board Member Sep 2022–Sep 2024

- Oversaw SURover team operations, leadership coordination, and project continuity within the chapter

Sabancı University SIAM Student Chapter | Financial Affairs Coordinator Sep 2022–Jun 2024

- Managed budgets and sponsorships; organized seminars, reading groups, and on-campus math competitions

Civic Involvement Projects (CIP) | Volunteer Apr 2021–Jun 2021

- Narrated and edited audiobooks for visually impaired listeners to improve accessibility

REFERENCES

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